

CoRoT-2b

R-0078

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_240419 · created 2026-07-15T20:17:41+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460419.9532 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.042 +/- 0.064
Transit depth (Rp/Rs)^2	0.18 +/- 0.54 [%]
Orbital Inclination (inc)	86.82 +/- 2.09
Airmass coefficient 1 (a1)	1.86 +/- 0.99
Airmass coefficient 2 (a2)	-0.32 +/- 0.74
Scatter in the residuals of the lightcurve fit is	53.96 %
Best Comparison Star	#1 - [212, 366]
Optimal Aperture	2.76
Optimal Annulus	4.75
Transit Duration (day)	0.0787 +/- 0.0097

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.042 ± 0.064 vs 0.1667 (deviation 1.95σ)
Duration fit vs published	0.0787 d vs 0.095 d (deviation 1.68σ)
Mid-time O–C	9.2 min
Depth SNR	0.7
Residual scatter	53.96 %

Light curve

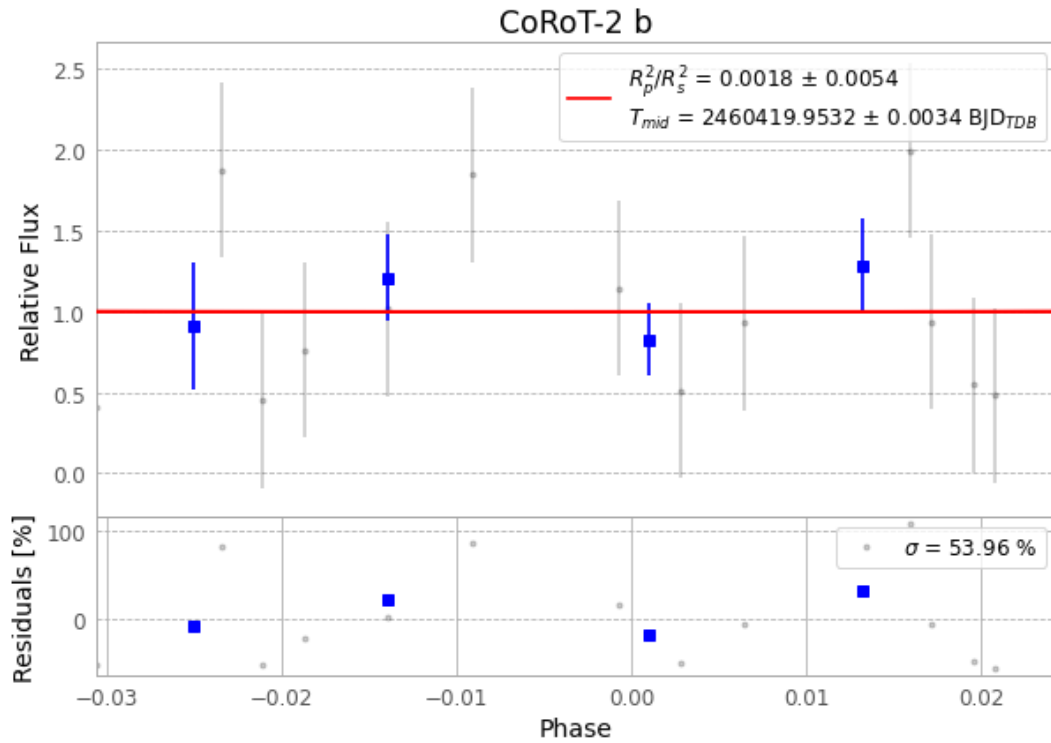


Figure 1 — FinalLightCurve_CoRoT-2 b_2024-04-19.png (SHA-256 6dacb49b5ce18e73...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [166, 314], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2a26d3c17dc69123...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

51 FITS frames (33 MB), CoRoT-2240419092416.FITS → CoRoT-2240419115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-240419.log (98577385b9d7...), FinalLightCurve_CoRoT-2 b_2024-04-19.png (6dacb49b5ce1...), FinalLightCurve_CoRoT-2 b_2024-04-19.csv (7280059198b0...)

Compute resources

Wall time 195.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 20:17Z.